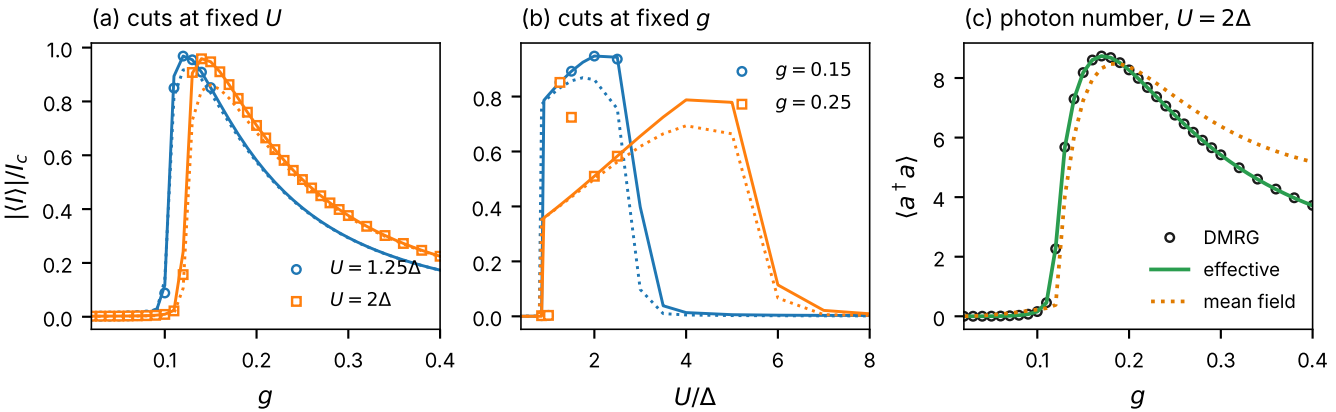
(a) cuts at fixed U
$|\langle I \rangle|/I_c$
1.0
0.8
0.6
0.4
0.2
0.0
0.1
0.2
0.3
0.4
g
$U = 1.25\Delta$
$U = 2\Delta$
(b) cuts at fixed g
$g = 0.15$
$g = 0.25$
0.8
0.6
0.4
0.2
0.0
2
4
6
8
U/Δ
(c) photon number, $U = 2\Delta$
$\langle a^\dagger a \rangle$
8
6
4
2
0
0.1
0.2
0.3
0.4
g
DMRG
effective
mean field